"Calculus 3"

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 21

1

Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.2 (line integrals)

- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

2

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3

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The Fundamental Theorem for Line Integrals

4

The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where F is any antiderivative of f, that is, a function F such that $F' = f$.

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

5

The Fundamental Theorem for Line Integrals

2 Theorem

Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leqslant t \leqslant b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C. Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

6

The Fundamental Theorem for Line Integrals: Proof

7

The Fundamental Theorem for Line Integrals: Proof

$$\begin{aligned}\int_C \nabla f \cdot d\mathbf{r} &= \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt \\ &= \int_a^b \left(\frac{\partial f}{\partial x}\,\frac{dx}{dt} + \frac{\partial f}{\partial y}\,\frac{dy}{dt} + \frac{\partial f}{\partial z}\,\frac{dz}{dt}\right) dt \\ &= \int_a^b \frac{d}{dt}\, f(\mathbf{r}(t))\, dt \qquad \text{(by the Chain Rule)} \\ &= f(\mathbf{r}(b)) - f(\mathbf{r}(a))\end{aligned}$$

8

Example: Let $f(x,y) = x^2e^y$. Evaluate $\int_C \nabla f \cdot dr$ where C is the curve parametrized by $r(t) = (2t, t^3)$, $1 \leq t \leq 2$.

9

[Extra Space]

Example: Let $f(x,y) = x^2e^y$. Evaluate $\int_C \nabla f \cdot dr$ where C is the curve parametrized by $r(t) = (2t, t^3)$, $1 \leq t \leq 2$.

10

Example: Let F$(x, y, z) = (yz, xz, xy)$. Evaluate $\int_C \mathrm{F} \cdot dr$ where C is the line segment from (1,0,-2) to (4,6,3).

11

[Extra Space]

Example: Let F$(x, y, z) = (yz, xz, xy)$. Evaluate $\int_C \mathrm{F} \cdot dr$ where C is the line segment from (1,0,-2) to (4,6,3).

12

Conservative Vector Fields

A vector field $\mathbf{F}$ is called a **conservative vector field** if it is the gradient of some scalar function, that is, if there exists a function f such that $\mathbf{F} = \nabla f$. In this situation f is called a **potential function** for $\mathbf{F}$.

Example: Let F$(x, y, z) = (yz, xz, xy)$. Is F conservative?

Example: Let G$(x, y, z) = (xy^2 + 3x, x^2y)$. Is G conservative?

13

Example: Let G$(x, y, z) = (xy^2 + 3x, x^2y)$. Is G conservative?

14

[Extra Space]

Example: Let G$(x, y, z) = (xy^2 + 3x, x^2 y)$. Is G conservative?

15

Recall!

Example: Let $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3}\,\vec{x}$ be a gravitational vector field (in 3D), and let $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$. Show that $F = \nabla f$.

16

Example: Evaluate $\int_C 2xdx + 2ydy$ where C is the curve parametrized by
$r(t) = \left(t^2 - 1, \sqrt{t+1}\right),\ 0 \le t \le 3.$

$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz \qquad \text{where } \mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$

17

[Extra Space]

Example: Evaluate $\int_C 2xdx + 2ydy$ where C is the curve parametrized by
$r(t) = \left(t^2 - 1, \sqrt{t+1}\right),\ 0 \le t \le 3.$

18

Path Independence

The line integrals of a vector field F are called path-independent if for any two paths C, D starting and ending at the same point we have

$$\int_C F \cdot dr = \int_D F \cdot dr.$$

Note: If F is conservative, then it is path-independent, by the Fundamental Theorem of Line Integrals.

Example: Let G$(x, y, z) = (xy^2 + 3x, x^2 y)$. Then G is path-independent.

19

Path Independence

The line integrals of a vector field F are called path-independent if for any two paths C_1, C_2 starting and ending at the same point we have

$$\int_{C_1} F \cdot dr = \int_{C_2} F \cdot dr.$$

A parametrized curve is called closed or a loop if it starts and ends at the same point.

Example: Let $r(t) = (\cos(t), \sin(t)),\ 0 \le t \le 2\pi$.
Then r(t) is a loop.

3 Theorem

$\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D if and only if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in D.

20

Example: Evaluate $\int_C F \cdot dr$ where C is the ellipse $4x^2 + 9y^2 = 1$ and
$\mathrm{F} = \left((1+xy)e^{xy}, x^2e^{xy}\right)$.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz \qquad \text{where } \mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$$

21

[Extra Space]

Example: Evaluate $\int_C F \cdot dr$ where C is the ellipse $4x^2 + 9y^2 = 1$ and
$F(x,y) = \left((1+xy)e^{xy}, x^2e^{xy}\right)$.

22

Questions?

23

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24

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Green’s Theorem

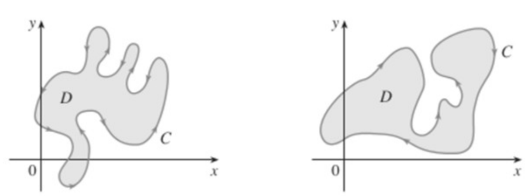
25

About Parametrized Curves

A parametrized curve is called closed or a loop if it starts and ends at the same point.

A closed curve is simple if its intersection with itself is at the start and the end, and nowhere else.

A simple closed curve on the plane is positively oriented if it goes counterclockwise. Otherwise it is negatively oriented.



(a) Positive orientation (b) Negative orientation

26

Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then

$$\int_C P\,dx + Q\,dy = \iint\limits_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

27

Example: Evaluate $\int_C x^4 dx + xy\, dy$ where C is the triangular curve along the segments from (0,0) to (1,0), from (1,0) to (0,1), and from (0,1) to (0,0).

28

[Extra Space]

Example: Evaluate $\int_C \ x^4 dx + xy\ dy$ where C is the triangular curve along the segments from (0,0) to (1,0), from (1,0) to (0,1), and from (0,1) to (0,0).

29

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Notes: Sometimes we write $\oint_C$ ... to denote we are integrating along a closed curve.

If F = (P,Q) is conservative, then $F = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$ and therefore

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}$$

30

Example: Evaluate $\oint_C \left(3y - e^{\sin(x)}\right)dx + \left(7x + \sqrt{y^4+1}\right)dy$ where C is the circle of radius 3 centered at the origin.

31

[Extra Space]

Example: Evaluate $\oint_C \left(3y - e^{\sin(x)}\right)dx + \left(7x + \sqrt{y^4+1}\right)dy$ where C is the circle of radius 3 centered at the origin.

32

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$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

$$P(x,y) = 0 \qquad P(x,y) = -y \qquad P(x,y) = -\frac{1}{2}y$$

$$Q(x,y) = x \qquad Q(x,y) = 0 \qquad Q(x,y) = \frac{1}{2}x$$

$$A = \oint_C x\,dy = -\oint_C y\,dx = \frac{1}{2}\oint_C x\,dy - y\,dx$$

33

Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

34

[Extra Space]

Example: Find the area of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

35

Example: Find the area of the triangle with vertices (0,0), (2,0), (2,3).

36

[Extra Space]

Example: Find the area of the triangle with vertices (0,0), (2,0), (2,3).

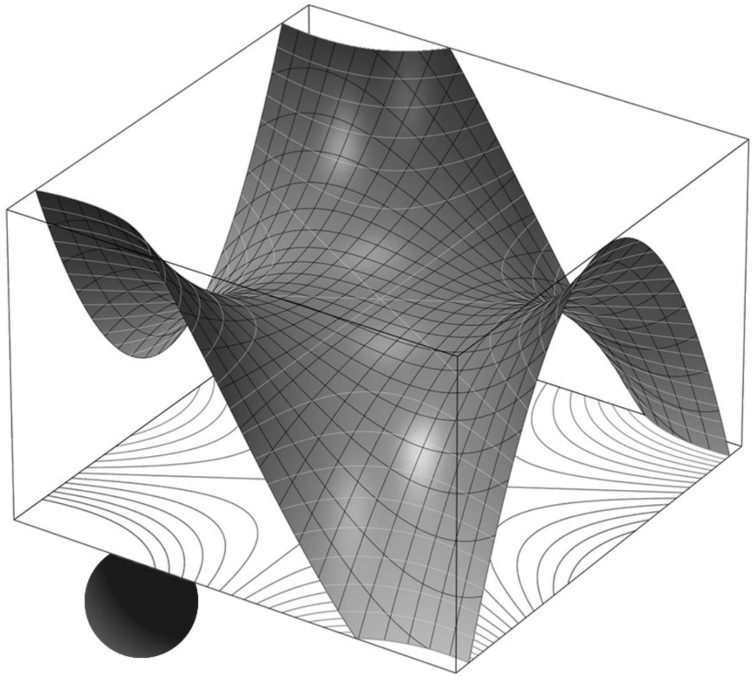
37

Questions?

38

Thank you

Until next time.



39